

Proof using Truth Tables:

P	Q	R	$P \wedge Q$	$Q \wedge R$	$P \wedge (Q \wedge R)$	$(P \wedge Q) \wedge R$
T	T	T	T	T	T	T
T	T	F	T	F	F	F
T	F	T	F	F	F	F
T	F	F	F	F	F	F
F	T	T	F	T	F	F
F	T	F	F	F	F	F
F	F	T	F	F	F	F
F	F	F	F	F	F	F

Fig 1

We can see from the truth table in Fig 1 that $P \wedge (Q \wedge R) \Leftrightarrow (P \wedge Q) \wedge R$ since they have the same corresponding truth values in their columns.

Let's Practice:

Prove:

$P \vee (Q \vee R) \Leftrightarrow (P \vee Q) \vee R$ using Truth tables

P	Q	R	$P \vee Q$	$Q \vee R$	$(P \vee Q) \vee R$	$P \vee (Q \vee R)$
T	T	T	T	T	T	T
T	T	F	T	T	T	T
T	F	T	T	T	T	T
T	F	F	T	F	T	T
F	T	T	T	T	T	T
F	T	F	T	T	T	T
F	F	T	T	T	T	T
F	F	F	F	F	F	F

$(P \vee Q) \vee R$ has the same truth value as $P \vee (Q \vee R)$

$$\therefore (P \vee Q) \vee R \Leftrightarrow P \vee (Q \vee R)$$

Proof using Semantics:

To prove:

$$P \wedge (Q \wedge R) \Leftrightarrow (P \wedge Q) \wedge R$$

To show that the two sides are equivalent, we need to prove:

$$P \wedge (Q \wedge R) \Rightarrow (P \wedge Q) \wedge R \text{ and}$$

$$(P \wedge Q) \wedge R \Rightarrow P \wedge (Q \wedge R)$$

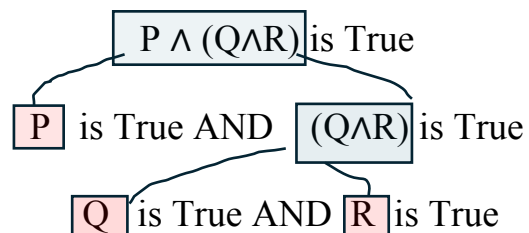
Let's start with:

$$P \wedge (Q \wedge R) \Rightarrow (P \wedge Q) \wedge R$$

In order to prove the above conditional statement, we need to show that

if $P \wedge (Q \wedge R)$ is True then $(P \wedge Q) \wedge R$ is also true.

Let's start with the assumption that $P \wedge (Q \wedge R)$ is True.



So, from our leaf nodes, we have **P is True AND Q is True AND R is True.**

Combining the results,

(P ∧ Q) is True AND R is True

Thus, $(P \wedge Q) \wedge R$ is True. So, we started with $P \wedge (Q \wedge R)$ as True and proved that

$(P \wedge Q) \wedge R$ is also True; Therefore,

$$P \wedge (Q \wedge R) \Rightarrow (P \wedge Q) \wedge R$$

Now, we need to Prove

$$(P \wedge Q) \wedge R \Rightarrow P \wedge (Q \wedge R)$$

To prove the above conditional statement, we need to show that

If $(P \wedge Q) \wedge R$ is True then $P \wedge (Q \wedge R)$ is also true.

Using similar arguments as above:

We start with **$(P \wedge Q) \wedge R$ as True.**

Which means **$(P \wedge Q)$ is True AND R is True.** $(P \wedge Q)$ is True makes **P True AND Q True.** Hence, we have **P True AND Q True AND R True.** Thus, **$(Q \wedge R)$ is True AND P is True.** Thus, **$P \wedge (Q \wedge R)$ is True.**

Since we have shown both conditional statements, we conclude the two sides are, indeed, logically equivalent.

Let's Practice:

Prove the following using Semantics:

$$P \vee (Q \vee R) \Leftrightarrow (P \vee Q) \vee R$$

2. Distributive Laws:

In arithmetic, we often come across the claim that *multiplication distributes across addition*:

$$\forall x, y, z \in R. x. (y + z) = x.y + x.z$$

Logical connectives, “ $\wedge$ ” and “ $\vee$ ” also distribute across one another:

Let P , Q , and R be logical statements.

Proof of $P \vee (Q \vee R) \Leftrightarrow (P \vee Q) \vee R$

To prove: 1) $P \vee (Q \vee R) \Rightarrow (P \vee Q) \vee R$

and

2) $(P \vee Q) \vee R \Rightarrow P \vee (Q \vee R)$

1) $P \vee (Q \vee R) \rightarrow \text{true}$

$\rightarrow P \rightarrow \text{true} \quad \text{OR}$

$(Q \vee R) \rightarrow \text{true}$

$\downarrow$

$Q \rightarrow \text{true} \quad \text{OR} \quad R \rightarrow \text{true}$

$\downarrow$

$(P \vee Q) \vee R$
 $\downarrow$
 true

$Q \rightarrow \text{true}$

$(P \vee Q) \rightarrow \text{true}$

$\downarrow$

$(P \vee Q) \vee R \rightarrow \text{true}$

$P \rightarrow \text{true}$

$(P \vee Q) \rightarrow \text{true}$

$\downarrow$

$(P \vee Q) \vee R \rightarrow \text{true}$

2) $(P \vee Q) \vee R \rightarrow \text{true}$

$(P \vee Q) \rightarrow \text{true} \quad \text{OR}$

$\downarrow$

$P \rightarrow \text{true} \quad \text{OR} \quad Q \rightarrow \text{true}$

$\downarrow$

$P \vee (Q \vee R) \rightarrow \text{true}$

$\downarrow$

$(Q \vee R) \rightarrow \text{true}$

$\downarrow$

$P \vee (Q \vee R) \rightarrow \text{true}$

$R \rightarrow \text{true}$

$\downarrow$

$(Q \vee R) \rightarrow \text{true}$

$\downarrow$

$P \vee (Q \vee R) \rightarrow \text{true}$

Then

$$P \wedge (Q \vee R) \Leftrightarrow (P \wedge Q) \vee (P \wedge R)$$

$$P \vee (Q \wedge R) \Leftrightarrow (P \vee Q) \wedge (P \vee R)$$

Proof using Truth Tables:

P	Q	R	$Q \vee R$	$P \wedge Q$	$P \wedge R$	$P \wedge (Q \vee R)$	$(P \wedge Q) \vee (P \wedge R)$
T	T	T	T	T	T	T	T
T	T	F	T	T	F	T	T
T	F	T	T	F	T	T	T
T	F	F	F	F	F	F	F
F	T	T	T	F	F	F	F
F	T	F	T	F	F	F	F
F	F	T	T	F	F	F	F
F	F	F	F	F	F	F	F

Fig 2

We can see from the truth table above that **$P \wedge (Q \vee R)$ and $(P \wedge Q) \vee (P \wedge R)$ have the same corresponding Truth values in their columns** thus they are logically equivalent.

Let's Practice:

Prove using Truth Tables:

$$P \vee (Q \wedge R) \Leftrightarrow (P \vee Q) \wedge (P \vee R)$$

(Proof of the above statements using semantics will be a part of homework!)

3. De Morgan's Laws

We shall now transition towards proving logical equivalences involving negations. The following law that we will discuss is named for the British mathematician *Augustus De Morgan*. He is also known for introducing the term *Mathematical Induction*!

According to De Morgan's Law:

Let P and Q be logical statements. Then

$$\neg (P \wedge Q) \Leftrightarrow \neg P \vee \neg Q$$

And

$$\neg (P \vee Q) \Leftrightarrow \neg P \wedge \neg Q$$

Proof:

We will prove the above claims using Truth Tables:

P	$\neg P$	Q	$\neg Q$	$P \wedge Q$	$\neg(P \wedge Q)$	$\neg P \vee \neg Q$
T	F	T	F	T	F	F
T	F	F	T	F	T	T
F	T	T	F	F	T	T
F	T	F	T	F	T	T

Fig 3

P	$\neg P$	Q	$\neg Q$	$P \vee Q$	$\neg(P \vee Q)$	$\neg P \wedge \neg Q$
T	F	T	F	T	F	F
T	F	F	T	T	F	F
F	T	T	F	T	F	F
F	T	F	T	F	T	T

Fig 4

We can see from the Truth Table in Fig 3 that $\neg(P \wedge Q)$ and $\neg P \vee \neg Q$ have the same truth values hence they are logically equivalent.

Similarly, from Fig4 we can conclude that $\neg(P \vee Q)$ and $\neg P \wedge \neg Q$ are logically equivalent.

Negation of conditional statements:

Consider a claim of the form $P \Rightarrow Q$. How do we negate such a statement? We need to find the exact cases for which $P \Rightarrow Q$ is not True. We know that $P \Rightarrow Q$ is True when either $\neg P$ is True, or Q is True.

That is the same as saying:

$$P \Rightarrow Q \Leftrightarrow \neg P \vee Q$$

Therefore,

$$\neg(P \Rightarrow Q) \Leftrightarrow \neg(\neg P \vee Q) \Leftrightarrow P \wedge \neg Q$$

We can use **De Morgan's law** to prove this :

$$\neg(P \Rightarrow Q) \Leftrightarrow \neg(\neg P \vee Q)$$

$$\Leftrightarrow \neg(\neg P) \wedge \neg Q$$

$$\Leftrightarrow P \wedge \neg Q$$

Let's Practice:

$P \wedge \neg Q$

Negate the following Statements:

1. $\forall x \in \mathbb{R} . x < 0 \vee x > 0$ $\exists x \in \mathbb{R} . x \geq 0 \wedge x \leq 0$
2. $\exists x \in \mathbb{N} . x < 10 \wedge x > 5$ $\forall x \in \mathbb{N} . x \geq 10 \vee x \leq 5$
3. $\exists x \in \mathbb{R} . \forall y \in \mathbb{R} . x > y \Rightarrow x^2 > y^2$
 $\forall x \in \mathbb{R} . \exists y \in \mathbb{R} . x > y \wedge x^2 \leq y^2$

$$\neg(x > y \Rightarrow x^2 > y^2)$$

$$\frac{P}{Q}$$

$$P \wedge \neg Q$$

$$x > y \wedge x^2 \leq y^2$$

Some extra Practice for fun 😊

Prove De Morgan's Law using semantics:

1. $\neg(P \wedge Q) \Leftrightarrow \neg P \vee \neg Q$

2. $\neg(P \vee Q) \Leftrightarrow \neg P \wedge \neg Q$

$$1. \quad \neg(P \wedge Q) \Leftrightarrow \neg P \vee \neg Q$$

1. To prove:

$$(i) \quad \neg(P \wedge Q) \Rightarrow \neg P \vee \neg Q$$

and

$$(ii) \quad \neg P \vee \neg Q \Rightarrow \neg(P \wedge Q)$$

(i) Assuming $\neg(P \wedge Q)$ is True

$$\begin{array}{c} \downarrow \\ P \wedge Q \text{ is false} \\ \downarrow \\ P \rightarrow \text{false} \text{ OR } Q \rightarrow \text{false} \\ \downarrow \qquad \qquad \downarrow \\ \neg P \rightarrow \text{True} \qquad \neg Q \rightarrow \text{True} \\ \downarrow \qquad \qquad \downarrow \\ \neg P \vee \neg Q \rightarrow \text{True} \qquad \neg P \vee \neg Q \rightarrow \text{True} \end{array}$$

(ii) Assuming $\neg P \vee \neg Q \rightarrow \text{True}$

$$\begin{array}{c} \downarrow \qquad \qquad \text{OR} \qquad \qquad \downarrow \\ \neg P \rightarrow \text{True} \qquad \neg Q \rightarrow \text{True} \\ \downarrow \qquad \qquad \downarrow \\ P \rightarrow \text{false} \qquad Q \rightarrow \text{false} \\ \downarrow \qquad \qquad \downarrow \\ P \wedge Q \rightarrow \text{false} \qquad P \wedge Q \rightarrow \text{false} \\ \downarrow \qquad \qquad \downarrow \\ \neg(P \wedge Q) \rightarrow \text{True} \qquad \neg(P \wedge Q) \rightarrow \text{True} \end{array}$$

$$2. (i) \neg(P \vee Q) \Rightarrow \neg P \wedge \neg Q$$

(To prove: $\neg(P \vee Q) \Leftrightarrow \neg P \wedge \neg Q$)

$$(ii) \neg P \wedge \neg Q \Rightarrow \neg(P \vee Q)$$

